

Practice 1

Part I. Single Choice (2 points each)

1. Which function is the entry point of every C++ program?

- A. start()
- B. run()
- C. main()
- D. init()

Answer:

2. Which object is used to display output on the screen?

- A. cin
- B. cout
- C. int
- D. return

Answer:

3. Which variable declaration is valid?

- A. int 3score;
- B. int score;
- C. int score!;
- D. int class;

Answer:

4. Which data type is most appropriate for storing a person's age?

- A. string
- B. char
- C. int
- D. bool

Answer:

5. Which statement both declares and initializes a variable named `value` to 100?

- A. int value;
- B. value = 100;

C. `int value =100;`

D. `100 = value;`

Answer: C

Part II. Fill in the Blank (2 points each)

6. The object used to print output to the screen is: _____

Answer:

7. Complete the code to declare an integer variable named `score` :

_____ `score;`

Answer:

8. Complete the code to declare and initialize a double precision floating point variable: _____`price =19.99;`

Answer:

9. The C++ data type used to store `true` or `false` values is:

Answer:

10. What is the output of the following code?

```
int x =5;
cout << x;
```

Output:

Answer:

Part III. True or False (2 points each)

11. A variable name can begin with a number.

Answer: False

12. A variable should be declared before it is used.

Answer: True

13. The following statement is valid C++ code:

```
char grade ='A';
```

Answer: True

14. The following statement stores the value 18.5 exactly:

```
int age =18.5;
```

Answer: False

15. The value stored in a variable can change while the program is running.

Answer: True

Part IV. Code Reading (5 points each)

16. What is the output of this program?

```
#include <iostream>
using namespace std;

int main() {
    int x = 10;
    cout << x;
    return 0;
}
```

Output:

Answer:

17. What is the output of this program?

```
#include <iostream>
using namespace std;

int main() {
    int a = 3;
    int b = 5;

    cout << a;
    cout << b;
```

```
    return 0;  
}
```

Output:

Answer:

Part V. Short Programming Questions (10 points each)

18. Print a Message

Write a C++ program that prints:

```
My name is Alex.  
I am learning C++.
```

19. Variable Practice

Write a C++ program that:

1. Declares an integer variable named `age`
2. Assigns it the value `18`
3. Prints:

```
Age = 18
```

Challenge Question (+10 Bonus Points)

20. Study the following code and write the output.

```
#include <iostream>  
using namespace std;  
  
int main() {  
    int x = 5;  
  
    cout << x << endl;  
  
    x = 8;  
  
    cout << x << endl;  
}
```

```
    return 0;  
}
```

Output:

Then answer: **Why are the two outputs different?**
